

assignment2 - Ohm's law, simulation

 This is a preview of the published version of the quiz

Started: Mar 22 at 7:03pm

Quiz Instructions

2 attempts.

You can use: https://phet.colorado.edu/sims/html/circuit-construction-kit-dc/latest/circuit-construction-kit-dc_en.html  (https://phet.colorado.edu/sims/html/circuit-construction-kit-dc/latest/circuit-construction-kit-dc_en.html)

Do not look at the hints

You can start and wait for us to cover circuit in series



Question 1 5 pts

The resistance of a filament that carries 2 A when a 10-V potential difference across it is

☐

10 ohms.

☐

more than 20 ohms.

☐

5 ohms.

☐

20 ohms.

☐

2 ohms.



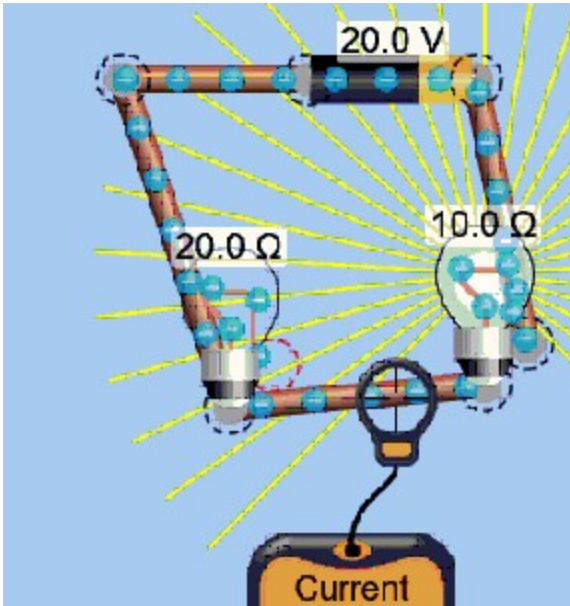
Question 2 10 pts

Two lamps with different filament thicknesses, and therefore different resistances, are connected in series.

What is true

hint: build a circuit with 2 light bulbs (different resistances) with the simulation. Measure the current

through each bulb.



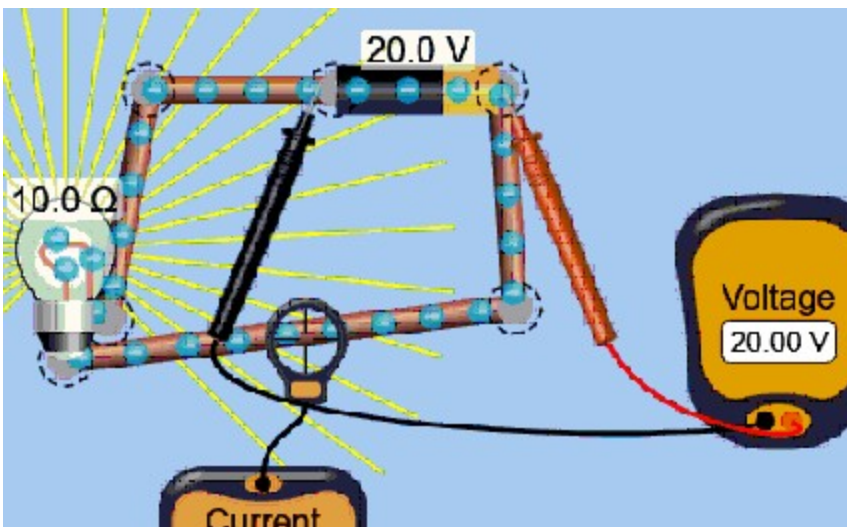
- ☐ The current through the thick filament is the largest
- ☐ The current through the bulbs is the same
- ☐ the current through the thin filament is the largest



Question 3 10 pts

If an electric toaster rated at 110 V is accidentally plugged into a 220-V outlet, the current drawn by the toaster will be (so same R but the voltage is $\times 2$)

hint: you can investigate by building the circuit below. Keep the resistance the same and see what happens to the current if the voltage of the power supply increases by a factor of 2 (from 20V to 40V for example)



- ☐ half its normal value.

☐

twice its normal value.

☐

the same as its normal value.

☐

none of the above



Question 4 5 pts

The current in two identical light bulbs connected in series is 0.25 A. The voltage across **both** bulbs is 110 V. The resistance of a single light bulb is

hint: The voltage across the 2 bulbs is 110V (from power supply). And the bulbs are identical (so same load, same resistance). The voltage across one bulb is therefore then apply Ohms law to find R.

☐

44 ohms.

☐

22 ohms.

☐

220 ohms.

☐

440 ohms.

☐

none of the above



Question 5 10 pts

The amount of current in an LED bulb rated 10.5W 120V is

☐

about 1 A.

☐

less than 1 A.

☐

more than 1 A.



Question 6 10 pts

A 25-W soldering iron runs on 110 V dc. What is its resistance?

hint: 1Kohms = 1000 ohms . Like 5K resistance is 5000 ohms.

☐

0.0020 Ω

2.8 k Ω 4.4 Ω 0.48 k Ω 

Question 7 10 pts

How much does it cost to operate a 25-W soldering iron for 8.0 hours if energy costs 8.0¢/kWh?



25¢



\$1.50



1.6¢



16¢



Question 8 10 pts

A 400-W computer (including its monitor) is turned on for 8.0 hours per day. If electricity costs 10¢ per kWh, how much does it cost to run the computer annually for a 365-day year?



\$15



\$150



\$1200



\$120



Question 9 5 pts

A 6.0-V battery that can store 500J of energy is connected to a resistor. How much charge must flow between the battery's terminals to completely drain the battery if it is fully charged?

Assume that the voltage of the battery remains the same until it is totally drained.



0.010 C



83 C

☐
3000 C

☐
0.07 C



Question 10 10 pts

A simple circuit has a total resistance of $30\ \Omega$. If a 2.0-A current is maintained in this circuit, how much energy is dissipated in this circuit in 4.0 seconds?

☐
4.8 J

☐
24 J

☐
48 J

☐
480 J

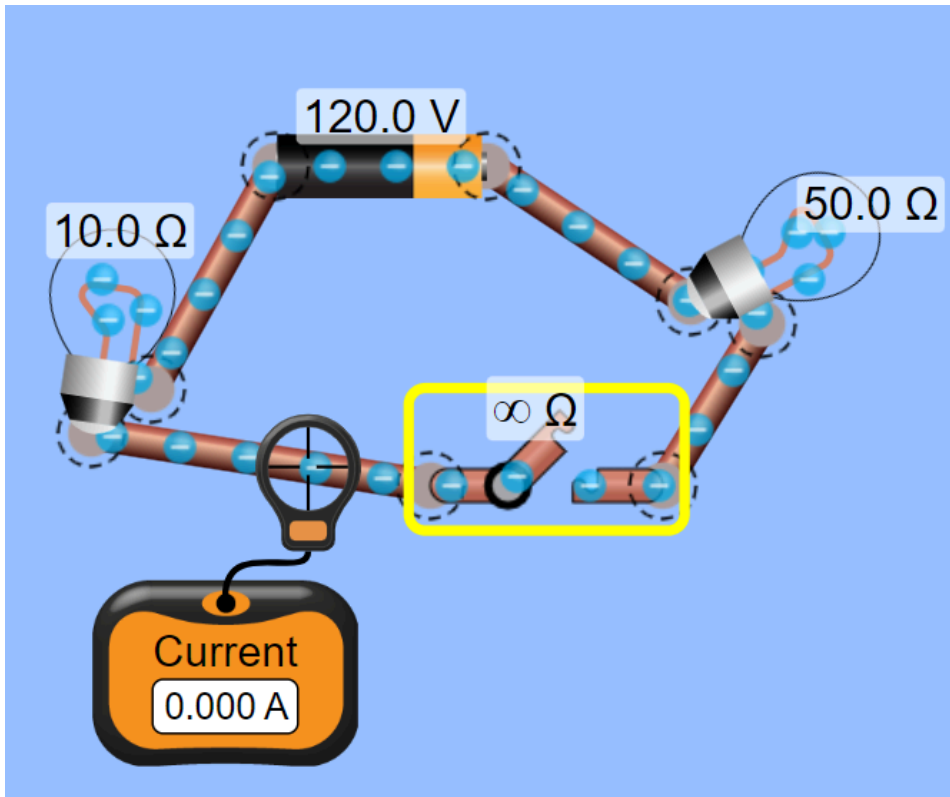
☐
6.0 J



Question 11 5 pts

https://phet.colorado.edu/sims/html/circuit-construction-kit-dc/latest/circuit-construction-kit-dc_en.html  (https://phet.colorado.edu/sims/html/circuit-construction-kit-dc/latest/circuit-construction-kit-dc_en.html)

Build the following circuit and close the switch. Which light bulb is the brightest (keep circuit for next question)


☐

They have the same brightness

☐

10ohms

☐

they are off

☐

50ohms

⋮

Question 12 5 pts

Building on the previous question. The current is

☐

0.5A

☐

2A

☐

1A

☐

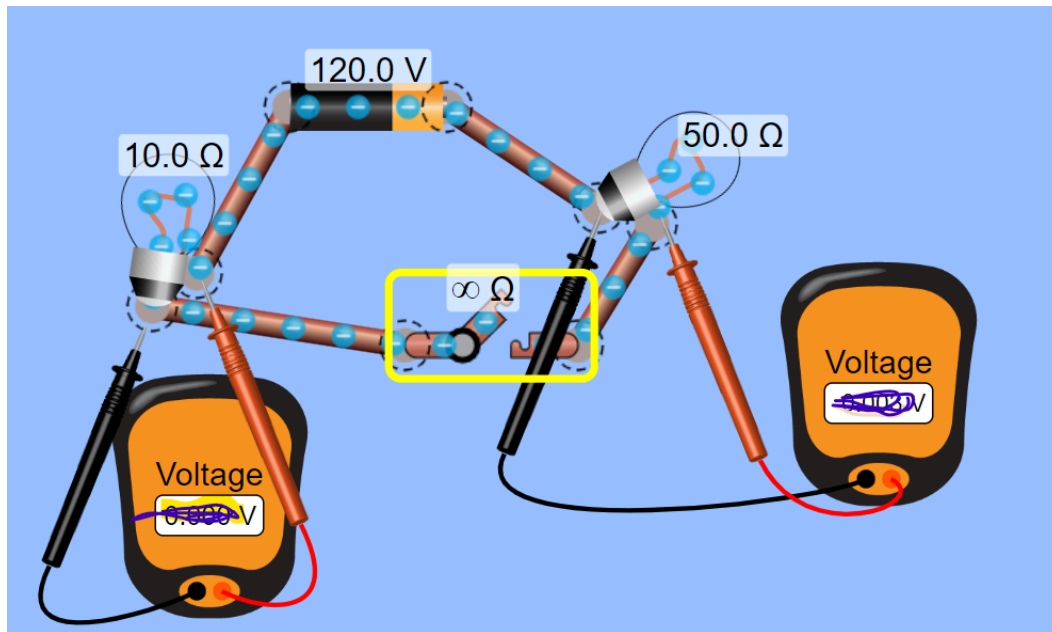
3V

⋮

Question 13 5 pts

Building on the previous question. Now get 2 voltmeter to measure the voltage across each load.

Close the switch. What is true



- ☐ the voltage across the 10 ohms is ten times smaller than the voltage across the 50 oms
- ☐ the voltage across the 50 ohms resistor is the same as the voltage across the 10 ohms
- ☐ The voltage across the 50 ohms is 10 times larger than the voltage across the 10ohms
- ☐ The voltage across the 50 ohms is 5 times the voltage across the 10 ohms

Not saved

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